

Document Effective Date
13/12/2021



QatarEnergy-IS PROCEDURE FOR LABORATORY SAFETY & CHEMICAL HYGIENE PLAN

DOC NO: IS-HSE-PRC-014

Rev: 01

Quality Reviewed By: OIH/14

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1.0 PURPOSE

The elements of this procedure are designed to minimize exposures of laboratory personnel to hazardous chemicals.

2.0 SCOPE AND APPLICABILITY

This Procedure applies to the laboratories in QatarEnergy-IS field.

3.0 ROLES AND RESPONSIBILITIES

OCCUPATIONAL HEALTH & PARAMEDIC (OHP)

- Develop, implement and revise this chemical hygiene plan, policies and practices. Maintain an awareness of current requirements concerning regulated substances.
- Assure appropriate records are maintained (training, audit reports, injury reports etc.).

OFFSHORE INSTALLATION SUPERINTENDENT (OIS)

- Ensure that facilities and training for use of any new materials being ordered are adequate.
- Ensure that facility and laboratory staff know and follow this Lab Safety and Chemical hygiene Procedure.
- Ensure regular, formal laboratory safety and housekeeping inspections, including routine inspections of any emergency equipment and PPE required by this plan are carried out.

LABORATORY CHEMIST

- Ensure purchasing, use, and disposal of chemicals and biological materials used in the laboratories are monitored in accordance with QatarEnergy-IS Procedure for Hazard Communication (IS-HSE-PRC-006) requirements and the QatarEnergy-IS Waste Management Procedure (IS-HSE-PRC-012).
- Plan and conduct each activity in accordance with this plan and other laboratory Procedures.
- Determine the appropriate levels of protective apparel and equipment needed in the laboratory. Ensure that the protective equipment is available, in working order and used.
- Perform a quality check on the ventilation every three months. Keep a log in the laboratory.
- Ensure that the anemometer used for the ventilation checks is maintained and calibrated to manufacturer's requirements.
- Develop and practice good personal chemical and biological hygiene habits.

SAFETY OFFICER

- Carry out weekly inspections of the laboratories to ensure good housekeeping, materials are stored and handled properly.

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- Ensure that correct PPE is being used per the requirements of the Material Safety Data Sheet (MSDS).
- Ensure that fume hood air flow checks are being completed.

4.0 PROCEDURAL STEPS

4.1 ACCIDENTS AND SPILLS

In case of accidents or spillage, consult the Material Safety Data Sheet for specific emergency response.

In general cases:

- **Eye Contact:** Promptly flush eyes with cool water for a minimum period (15 minutes) and seek medical attention.
- **Ingestion:** Immediately seek medical attention.
- **Skin Contact:** Promptly flush the affected area with water for a minimum period (15 minutes) and remove any contaminated clothing. Seek medical attention.
- **Spills:** Contact the Facility Safety Officer to report any spillage and for emergency spill response.

4.2 AVOID "ROUTINE" EXPOSURE

- Do not smell or taste chemicals.
- Vent apparatus that may discharge toxic chemicals (vacuum pumps, distillation columns, etc.) into local exhaust devices.
- Inspect gloves before use.
- Wash areas of exposed skin well before leaving the laboratory.
- Do not use mouth suction when using a pipette or when starting a siphon.

4.3 CHEMICAL PURCHASING, DISTRIBUTION AND STORAGE

- Purchasing:** Each person ordering a substance that has a PEL, TLV, or is a carcinogen (see Hazardous Chemical List, Select Carcinogen List) shall contact the HSE Department to obtain information regarding the safe use and storage of the hazardous material or to recommend a less hazardous material.
- Stockroom/Storerooms:** Toxic substances should be segregated in a well-identified area with local exhaust ventilation. Flammable materials shall be stored in a flammable storage cabinet. Corrosive materials shall be stored in a dedicated corrosives cabinet. Acids and bases shall be separated. Stored chemicals in stockrooms shall be examined at least annually for replacement, deterioration and container integrity. Stockroom/storerooms shall not be used as preparation or repackaging areas.
- Distribution:** When chemicals are hand carried, secondary containment shall be practiced (the chemical shall be placed in a container).
- Laboratory storage:** Quantities of corrosive, flammable and toxic chemicals in the laboratories shall not exceed a one-day supply unless the

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materials are stored in flammable, corrosive and separated toxic cabinets with adequate ventilation.

4.4 EATING, DRINKING, SMOKING

Eating, drinking, smoking, handling contact lenses and applying cosmetics is strictly prohibited in areas where hazardous laboratory chemical or biological materials are used or are intended for use.

4.5 EQUIPMENT AND GLASSWARE

- Handle and store laboratory glassware with care to avoid damage.
- Inspect glassware before each use and do not use damaged glassware.
- Use extra care with evacuated glass apparatus. Shield or wrap them to contain chemicals and fragments should implosion occur.
- Use equipment only for its designed purpose.

4.6 HOUSEKEEPING, MAINTENANCE AND INSPECTIONS

- Floors shall be cleaned regularly.
- Keep the work area clean and uncluttered, with chemicals and equipment properly labelled and stored. Clean up the work area on completion of an operation or at the end of each day.
- The Facility Safety Officer shall conduct periodic inspections on all laboratories.
- Respirators shall be regularly inspected.
- Equipment taken out of service shall be disconnected from the power source and tagged 'Do not use', or alternatively, the circuit breaker may be removed to prevent accidental restarting of the equipment.

4.7 TRAINING

- The laboratory safety training programme shall be conducted as per the Training Matrix.
- The HSE Department shall regularly review the safety training status of laboratory staff.

4.8 PERSONAL PROTECTION

- Ensure that people, including visitors, wear appropriate eye protection where chemicals and biological agents are stored or handled.
- Wear appropriate gloves when the potential for contact with toxic materials or allergens exists. Inspect the gloves before each use and replace them periodically or when damaged.
- Use of low protein, non-powdered latex gloves is recommended to help reduce the risk of latex allergies. Contact the HSE Department if any lab personnel have an allergic reaction to latex gloves.
- Use appropriate respiratory equipment when air contaminant concentrations are not sufficiently restricted by engineering controls.

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- Fume hoods shall be used with any appreciably volatile substance with a TLV of less than 50 ppm.
- Use any other protective and emergency apparel and equipment as appropriate.
- Wear PPE as required QatarEnergy-IS Procedure for Personal Protective Equipment Selection and Use (IS-HSE-PRC-005).
- Avoid use of contact lenses in the laboratory unless necessary. If they are used, inform the supervisor so special precautions can be taken.
- Change laboratory coats or overalls immediately upon contamination and shall be discarded as hazardous waste.

4.9 PLANNING

Seek information and advice about hazards before starting an experiment. Plan the work activity using appropriate protective Procedures and positioning of equipment before beginning any new operation.

4.10 SIGNS AND LABELS

- Chemical waste containers shall be labelled using the Hazardous Waste labels as per the QatarEnergy-IS Waste Management Procedure (IS-HSE-PRC-012).
- Chemicals that are mixed as reagents for later experiments shall be labelled with the chemical name, concentration and hazard warning as per QatarEnergy-IS Hazard Communications Procedure (IS-HSE-PRC-006).
- Laboratories using hazardous chemicals shall have a sign at the entrance designating the area "Authorized Personnel Only" and "Chemical Hazard."
- The original manufacturers label shall be retained without alteration or mutilation on the container or replaced with an appropriate label if the contents are changed.
- Good laboratory practice ensures the dating of all chemical containers on arrival and upon opening.

4.11 UNATTENDED OPERATIONS

Leave lights on, place an appropriate sign on the door, and provide for containment of toxic substances in the event of failure of a utility service (such as cooling water) to an unattended operation.

4.12 USE OF FUME HOODS

- Use the fume hoods for operations that might result in release of toxic chemical vapours or dust.
- Use a fume hood or other local ventilation device when working with any appreciably volatile substance with a TLV of less than 50ppm.
- Confirm adequate fume hood performance before use. Keep the fume hood sash closed at all times except when adjustments within the fume hood are being made.

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- Perform a quality check on ventilation every three months by raising the sash 14" open and recording three readings: one from the center and two from sides of the hood at a distance of 1" from the sides. Average the three readings and ensure that the velocity into the hood should be adequately within the range of 60-100 linear feet per minute (ft/m).
- Keep materials stored in fume hood to a minimum and do not allow them to block vents or airflow. Hazardous materials should be used 6 inches within the sash of the fume hood.
- Leave the fume hood "on" when it is not in active use if toxic substances are stored in it.

4.13 WASTE DISPOSAL PROGRAM

- Ensure that the plan for each laboratory operation includes plans and training for waste disposal in compliance the QatarEnergy-IS Waste Management Procedure (IS-HSE-PRC-012).
- Deposit chemical waste in appropriately labelled receptacles.
- Do not discharge concentrated acids or bases; toxic, malodorous, or lachrymatory substances; or any oil containing materials into drains. Hazardous and Non-hazardous trash shall be segregated and disposed at the appropriate location.
- Radioisotopes shall be disposed of in accordance with the direction of the Radiation Protection Officer.

5.0 REFERENCES TO OTHER DOCUMENTS

TABLE 1: REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/External Document (if any)	Document Referencing
IS HSE Training Procedure	IS-HSE-PRC-002	Internal Document	Sideward
IS Procedure for Personal Protective Equipment Selection and Use	IS-HSE-PRC-005	Internal Document	Sideward
IS Hazard Communication Procedure	IS-HSE-PRC-006	Internal Document	Sideward
IS Waste Management Procedure	IS-HSE-PRC-012	Internal Document	Sideward
IS Respiratory Protection procedure	IS-HSE-PRC-013	Internal Document	Sideward
IS Industrial Hygiene Protection The Control of	IS-HSE-PRC-024	Internal Document	Sideward

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Exposure to Hazardous Substances			
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APPENDIX

APPENDIX A: FACILITY LABORATORY CHEMICALS LIST

LIST OF CHEMICALS USED ON OFFSHORE FACILITY LABORATORIES:

No.	Chemical Name
1	Acetic acid
2	Acetone
3	Ammonia
4	Ammonium Chloride
5	Ammonium Thiocyanate
6	Bacteria Growth Media PRD & GAB
7	Barium Chloride
8	Buffer Solutions (pH-4,7,9)
9	Calcium Bromide
10	Calcium Carbonate
11	Calcium Chloride
12	Chloroform
13	Citric Acid
14	Demulsifier F-46 (Knock Out)
15	Dissolved O2 Ampoules
16	DPD Free Chlorine Reagent
17	DPD Total Chlorine Reagent
18	EDT Disodium Salt
19	Eriochrome Black T
20	Ferric Chloride Hexahydrate
21	Hydrochloric Acid
22	n – Heptane
23	n – Hexane
24	Hydrogen Sulphide Chemets Ampoules
25	Iso Butyl Alcohol
26	Isoton Diluent II
27	Magnesium Chloride

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No.	Chemical Name
28	Methanol
29	Methyl Orange
30	Nitric Acid
31	Phenolphthalein
32	Potassium Chloride
33	Potassium Chromate
34	Potassium Hydrogen Phthalate
35	1,10 Phenanthroline
36	Silver Nitrate
37	Sodium Chloride
38	Sodium Hydroxide Pellets
39	Sodium Thiosulphate
40	Sulphuric Acid
41	Silicone Oil
42	Toluene
43	Xylene

APPENDIX B – REVISION HISTORY LOG**Revision Number: 01****Date: 06/11/2021**

Item Revised	Revision Description	Page No.
All	Document is updated based on new Management system documentation format and new QatarEnergy-IS organization structure	
Remarks:		

APPENDIX C – LIST OF ABBREVIATIONS

Term	Abbreviation
HEPA	High-efficiency particulate air-purifying
MSDS	Material Safety Data Sheets.
PEL	Permissible Exposure Limit
PPE	Personal Protective Equipment.
SCBA	Self-contained Breathing Apparatus
TLV	Threshold Limit Value